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Inventor(s)	Carrier; Philippe et al.

Extruder head assembly for additive manufacturing machines

Abstract

An extruder head assembly, which is for an additive manufacturing machine, includes a melt zone component, and a nozzle operatively connected to the melt zone component. The melt zone component is operatively connected to a cold end component for receiving and heating filament. The melt zone component includes a heat source, and an interior member in thermal communication with the heat source. The interior member defines therein a passage, at least one melt channel, and at least one aperture fluidly connecting the passage to the at least one melt channel. The passage and the at least one melt channel are fluidly connected downstream from the at least one aperture, the nozzle being configured to receive heated filament material from the passage and the at least one melt channel.

Inventors: Carrier; Philippe (Montreal, CA), Gelinas-Guy; Maxence (Boucherville, CA), Jasmin; Thomas (Montreal, CA)

Applicant: DYZE DESIGN (Lemoyne, CA)

Family ID: 93122739

Assignee: DYZE DESIGN (Lemoyne, CA)

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Primary Examiner: Sultana; Nahida

Attorney, Agent or Firm: BCF LLP

Background/Summary

TECHNICAL FIELD

(1) The present technology relates to extruder head assemblies for additive manufacturing machines.

BACKGROUND

(2) Additive manufacturing machines, also referred to as 3D printers, have extruder head assemblies for outputting fabrication material. Fabrication material, for example in the form of a filament, is received in the extruder head assembly, and is heated until melted. The melted material is then extruded from the extruder head assembly to manufacture an item.

(3) Conventional extruder head assemblies do present some inconveniences. The filament can heat up unevenly, with some sections of the filament reaching a desired temperature and/or melting before other sections. For instance, a core of the filament can be at a lower temperature than a radially outer portion of the filament. This can negatively impact extrusion thereof.

(4) Therefore, there is a desire for a technology that can overcome at least some of the above-described drawbacks.

SUMMARY

(5) It is an object of the present technology to ameliorate at least some of the inconveniences present in the prior art.

(6) According to one aspect of the present technology, there is provided an extruder head assembly for an additive manufacturing machine. The extruder head assembly includes a melt zone component and a nozzle. The melt zone component is operatively connected to a cold end component for receiving and heating a filament advanced by the cold end component. The nozzle is operatively connected to the melt zone component, and is configured to receive the filament having been heated from the melt zone component. The melt zone component includes a heat source and an interior member. The interior member, which is in thermal communication with the heat source, defines therein a passage, at least one melt channel, and at least one aperture fluidly connecting the passage to the at least one melt channel. The passage and the at least one melt channel are fluidly connected downstream from the at least one aperture, the nozzle being configured to receive heated filament material from the passage and the at least one melt channel.

(7) In some embodiments, the at least one aperture defines a central aperture axis that is generally perpendicular to a plane containing the at least one aperture, the passage defines a central passage axis extending along a length of the passage, and the central aperture axis is generally perpendicular to the central passage axis.

(8) In some embodiments, the at least one aperture defines a central aperture axis that is generally perpendicular to a plane containing the at least one aperture, the passage defines a central passage axis extending along a length of the passage, and the central aperture axis is skewed relative to the central passage axis.

(9) In some embodiments, the at least one melt channel defines a central melt axis extending along the length of the at least one melt channel; and the central melt axis is parallel to the central passage axis.

(10) In some embodiments, the passage defines a central passage axis extending along a length of the passage; the at least one melt channel defines a melt channel axis extending along a length of the at least one melt channel; and the melt channel axis is parallel to the central passage axis.

(11) In some embodiments, the passage has a first passage diameter upstream from the at least one aperture; a second passage diameter downstream from the at least one aperture; and the first passage diameter is greater than the second passage diameter.

(12) In some embodiments, the at least one aperture includes a first aperture and a second aperture,

the second aperture being spaced from the first aperture along a length of the interior member.

(13) In some embodiments, the first aperture has a first aperture diameter, the second aperture has a second aperture diameter, and the first aperture diameter is greater than the second aperture diameter.

(14) In some embodiments, the passage has a first passage diameter upstream from the first aperture; a second passage diameter downstream from the first aperture and upstream from the second aperture; and a third passage diameter downstream from the second aperture. The first passage diameter is greater than the second passage diameter and the third passage diameter, and the second passage diameter is greater than the third passage diameter.

(15) In some embodiments, the at least one melt channel includes a first melt channel and a second melt channel.

(16) In some embodiments, the at least one aperture fluidly connects the passage to the first and second melt channels.

(17) In some embodiments, the at least one aperture includes a first aperture fluidly connecting the passage to the first melt channel, and a second aperture fluidly connecting the passage to the second melt channel.

(18) In some embodiments, the at least one melt channel includes a first melt channel, a second melt channel and a third melt channel. The at least one aperture includes a first aperture fluidly connecting the passage to the first melt channel, a second aperture fluidly connecting the passage to the second melt channel, and a third aperture fluidly connecting the passage to the third melt channel.

(19) In some embodiments, the at least one aperture further includes a fourth aperture fluidly connecting the passage to the first melt channel, the fourth aperture being spaced from the first aperture along a length of the interior member.

(20) In some embodiments, the first aperture has a first aperture diameter, the fourth aperture has a second aperture diameter, and the first aperture diameter is greater than the second aperture diameter.

(21) In some embodiments, a first passage diameter upstream from the first aperture; a second passage diameter downstream from the first aperture and upstream from the fourth aperture; and a third passage diameter downstream from the fourth aperture. The first passage diameter is greater than the second passage diameter and the third passage diameter, and the second passage diameter is greater than the third passage diameter.

(22) In some embodiments, the at least one aperture is configured to direct melted filament material toward the at least one melt channel, and the passage is configured to heat unmelted material of the filament.

(23) In some embodiments, the interior member has a generally cylindrical form.

(24) In some embodiments, the melt zone component further includes a holding member selectively receiving the interior member therein, one end of the holding member being selectively connected to the cold end component, and another end of the holding member being selectively connected to the nozzle.

(25) In some embodiments, the extruder head assembly includes a thermal block, and the one end of the holding member is selectively connected to the cold end component by the thermal block.

(26) In some embodiments, the heat source includes a heat transfer member, and the holding member is disposed in the heat transfer member.

(27) In some embodiments, the heat source is a serpentine heater surrounds the holding member.

(28) In some embodiments, the extruder head assembly further includes the cold end component for receiving the filament, the cold end component including filament advancing members.

(29) According to another aspect of the present technology, there is provided an additive manufacturing machine including a frame; a printer bed supported by the frame; the extruder head assembly according to the above aspect or according to the above aspect and one or more of the

above embodiments; and a controller supported by the frame, and operatively connected to the extruder head assembly.

(30) According to another aspect of the present technology, there is provided an internal member for an extruder head assembly for an additive manufacturing machine. The internal member defines a passage, at least one melt channel, and at least one aperture fluidly connecting the passage to the at least one melt channel. The passage and the at least one melt channel connecting to form a nozzle passage configured to fluidly connect to a nozzle.

(31) In the context of the present specification, unless expressly provided otherwise, the words “first”, “second”, “third”, etc. have been used as adjectives only for the purpose of allowing for distinction between the nouns that they modify from one another, and not for the purpose of describing any particular relationship between those nouns.

(32) It must be noted that, as used in this specification and the appended claims, the singular form “a”, “an” and “the” include plural referents unless the context clearly dictates otherwise.

(33) As used herein, the term “about” in the context of a given value or range refers to a value or range that is within 20%, preferably within 10%, and more preferably within 5% of the given value or range.

(34) As used herein, the term “and/or” is to be taken as specific disclosure of each of the two specified features or components with or without the other. For example “A and/or B” is to be taken as specific disclosure of each of (i) A, (ii) B and (iii) A and B, just as if each is set out individually herein.

(35) For purposes of the present application, terms related to spatial orientation when referring to an additive manufacturing machine and components in relation to the additive manufacturing machine, such as “vertical”, “horizontal”, “forwardly”, “rearwardly”, “left”, “right”, “above” and “below”, are as they would be understood by a person standing in front of the additive manufacturing machine.

(36) Embodiments of the present technology each have at least one of the above-mentioned object and/or aspects, but do not necessarily have all of them. It should be understood that some aspects of the present technology that have resulted from attempting to attain the above-mentioned object may not satisfy this object and/or may satisfy other objects not specifically recited herein.

(37) Additional and/or alternative features, aspects, and advantages of embodiments of the present technology will become apparent from the following description, the accompanying drawings, and the appended claims.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) For a better understanding of the present technology, as well as other aspects and further features thereof, reference is made to the following description which is to be used in conjunction with the accompanying drawings, where:

(2) FIG. 1 is a perspective view taken from a top, front, left side of an additive manufacturing machine having an extruder head assembly according to an embodiment of the present technology;

(3) FIG. 2 is a perspective view taken from a top, front, left side of the extruder head assembly of FIG. 1;

(4) FIG. 3 is a top plan view of the extruder head assembly of FIG. 1;

(5) FIG. 4 is a cross-sectional view of the extruder head assembly of FIG. 1, taken along line 4-4 of FIG. 3;

(6) FIG. 5A is a top plan view of a melt zone component and nozzle of the extruder head assembly of FIG. 1;

(7) FIG. 5B is a cross-sectional view of the melt zone component and nozzle of FIG. 5A, taken

along the line 5B-5B of FIG. 5A;

(8) FIG. 5C is a cross-sectional view of the melt zone component and nozzle of FIG. 5A, taken along the line 5C-5C of FIG. 5A;

(9) FIG. 6 is a perspective view taken from a bottom, rear, right side of an interior member of the melt zone component of FIG. 5A;

(10) FIG. 7A is a side elevation view of the interior member of FIG. 6;

(11) FIG. 7B is a bottom plan view of the interior member of FIG. 6;

(12) FIG. 7C is a cross-sectional of the interior member of FIG. 6, taken along line 7C-7C of FIG. 7A;

(13) FIG. 8 is a perspective view of a melt zone component according to an alternative embodiment of the present technology;

(14) FIG. 9 is a bottom, side perspective view of an interior member according to an alternative embodiment of the present technology;

(15) FIG. 10A is a side elevation view of the interior member of FIG. 9;

(16) FIG. 10B is a bottom plan view of the interior member of FIG. 9; and

(17) FIG. 10C is a cross-sectional view of part of the interior member of FIG. 9, taken along line 10C-10C of FIG. 10B.

(18) Unless otherwise noted, the Figures may not be drawn to scale.

DETAILED DESCRIPTION

(19) The present disclosure is not limited in its application to the details of construction and the arrangement of components set forth in the following description or illustrated in the drawings. The disclosure is capable of other embodiments and of being practiced or of being carried out in various ways. Also, the phraseology and terminology used herein is for the purpose of description and should not be regarded as limiting. The use of “including”, “comprising”, or “having”, “containing”, “involving” and variations thereof herein, is meant to encompass the items listed thereafter as well as, optionally, additional items. In the following description, the same numerical references refer to similar elements.

(20) The present technology relates to extruder head assemblies for additive manufacturing machines. An extruder head assembly according to the present technology is configured to improve uniformity in temperature and material state in additive manufacturing material at the extrusion point of the extruder head assembly.

(21) Referring to FIG. 1, an additive manufacturing machine 50 is shown according to one non-limiting embodiment of the present technology. More specifically, the additive manufacturing machine 50 is a three-dimensional printer (3D printer) 50. The 3D printer 50 includes a frame 52, a printer bed 54 supported by the frame 52, an extruder head assembly 56 operatively connected to the frame 52, and a controller 58 (schematically shown in FIG. 1) supported by the frame 52 and operatively connected to the extruder head assembly 56.

(22) The frame 52 has a base 60, a front wall 62, side walls 64, a rear wall 66, and a top wall 68. The base 60 and the front, side, rear and top walls 60, 62, 64, 66 and 68 define an enclosed printing space. The front wall 62 has a window 63 which allows to look at the enclosed printing space. It is contemplated that in some embodiments, the window 63 could be omitted, and the front wall 62 could simply define an aperture. The top wall 68 defines an aperture 69, which enables, as will be described below, the extruder head assembly 56 to move relative to the frame 52. It is also contemplated that in some embodiments, the front, side, rear and top walls 60, 62, 64, 66 and 68 could be omitted, and the frame 52 could have columns disposed at each corner thereof. The frame 52 also includes a moveable member 70, which generally extends between the side walls 64. The moveable member 70 is moveable between the front and rear walls 60, 66 by actuators (not shown) which are communicatively connected to the controller 58. It is contemplated that in some embodiments, the frame 52 could include additional moveable members.

(23) The printer bed 54, as mentioned above, is supported by the frame 52. More specifically, the

printer bed **54** is moveably connected to the rear wall **66**. In some embodiments, the printer bed **54** could be moveably connected to the base **60**. Indeed, the printer bed **54**, which has a flat upper surface, is moveable in the vertical direction by actuators (not shown) that are communicatively connected to the controller **58**. It is contemplated that in some embodiments, the printer bed **54** could be generally fixed relative to the frame **52**. For instance, the printer bed **54** could be connected to the base **60** such that the printer bed **54** would only be vertically moveable to adjust a horizontal level thereof (e.g., raising or lowering a corner of the printer bed **54**). In such instances, the extruder head assembly **56** would be moveable in the vertical direction.

(24) The 3D printer **50** includes a filament spool **72** (only partially shown in FIG. **1**) that is connected to an external surface of the rear wall **66**. It is contemplated that the filament spool **72** could be connected elsewhere on the 3D printer **50**. The filament spool **72** holds the filament that is configured to be fed to the extruder head assembly **56**.

(25) The 3D printer further includes a filament tube **74** for guiding the filament from the filament spool **72** to the extruder head assembly **56**. Thus, one end of the filament tube **74** receives the filament from the filament spool **72**, and the other end of the filament tube **74** is connected to the extruder head assembly **56**. Another tube **75** is disposed adjacent to the filament tube **74**, and contains, inter alia, electrical wiring for powering the extruder head assembly **56**.

(26) The extruder head assembly **56** is configured to receive filament from the filament spool **72**, to melt the filament, and to extrude melted filament material to manufacture an object.

(27) The controller **58**, which is communicatively or operatively connected to a number of components of the 3D printer **450**, is configured to control various elements of the 3D printer **50**, including for example operation of the extruder head assembly **56**.

(28) With reference to FIGS. **2** to **4**, the extruder head assembly **56**, which includes a carriage **80**, a cold end component **82**, a melt zone component **84** and a nozzle **86**, will now be described in greater detail.

(29) The carriage **80** moveably connects the extruder head assembly **56** to the moveable member **70**. Indeed, the carriage **80**, which is operatively connected to the controller **58**, has actuators (not shown) enabling the carriage **80**, and thus the extruder head assembly **56**, to move along the moveable member **70** (i.e., between the side walls **64**).

(30) The cold end component **82**, which is connected to the carriage **80**, includes a housing **90** having a rear casing **92** and a front casing **94**. It is contemplated that in some embodiments, the housing **90** could be a unitary housing. The rear casing **92** is connected to the carriage **80**, and the front casing **94** is connected to the rear casing **92**.

(31) The cold end component **82** also includes a feeding tip **96** that is connected to the housing **90**. In the illustrated embodiment, the feeding tip **96** is disposed at a top of the front casing **94**. The feeding tip **96** is configured to connect to an end of the filament tube **74**, and thus guides the filament into the housing **90**. The cold end component **82** has a single feeding tip **96**, the extruder head assembly **56** being configured to have a single input filament.

(32) With reference to FIG. **4**, disposed within the housing **90**, the cold end component **82** includes filament advancing members **100** for driving the filament along the housing **90**. In the illustrated embodiment, there are six filament advancing members **100**: three filament advancing members **100a**, and three filament advancing members **100b** disposed opposite the members **100a**. It is understood that the number of filament advancing members **100a**, **100b** could vary from one embodiment to another. The filament advancing members **100a**, **100b** are rotationally connected to the housing **90**, and are operatively connected to a motor (not shown), which is operable by the controller **58**. In the present embodiment, the filament advancing members **100a**, **100b** are wheels which are configured to frictionally engage with the filament. In other embodiments, the filament advancing members **100a**, **100b** could be gears or other engaging members. The filament advancing members **100a** are spaced from the filament advancing members **100b** so as to provide sufficient clearance for the filament to fit therebetween. As will be described below, the filament

advancing members **100a**, **100b** are laterally moveable relative to one another.

(33) Referring to FIGS. **3** and **4**, the cold end component **82** also includes a knob **102** that is connected to the housing **90**, and that is operatively connected to the filament advancing members **100a**, **100b**. As a result of the operative connection between the knob **102** and the filament advancing members **100**, rotation of the knob **102** in one direction causes the filament advancing members **100a**, **100b** to move toward one another, and rotation of the knob **102** in the other direction causes the filament advancing members **100a**, **100b** to move away from one another. Rotating the knob **102**, and thus moving the filament advancing members **100a**, **100b** toward or away from one another, can be done so that the filament advancing members **100a**, **100b** can adequately grip onto the filament, and/or so that the filament advancing members **100a**, **100b** can grip filament of various sizes.

(34) The cold end component **82** also includes a fan **104** that is connected to the front casing **94**. The fan **104** can provide ventilation, which can assist in maintaining the cold end component **82** at a lower temperature than the melt zone component **84**.

(35) Connected to the bottom of the housing **90** is the melt zone component **84**, which includes a heat source **110**, a holding member **112**, an interior member **114**, and a thermal block **116**. The thermal block **116**, the holding member **112** and the interior member **114** are at least partially received in the heat source **110**, and the interior member **114** and the thermal block **116** are at least partially received in the holding member **112**.

(36) In the present embodiment, the heat source **110** includes a heat transfer member **110** and a heater **120** that is received in the heat transfer member **110**. Heat from the heater **120** is transferred generally uniformly to the holding member **112** by the heat transfer member **110**. The heat source **110** is configured and arranged to minimize heat transfer to the housing **90**, with aid from inter alia the thermal block **116**. It is contemplated that the heat source **110** could vary from one embodiment to another. For example, in an alternative embodiment of the present technology shown in FIG. **8**, the heat source **110** is a serpentine heater that surrounds the holding member **112**. Returning to FIG. **4**, the heat source **110** defines a receiving section **122** extending along an entire length thereof. The receiving section **122** is configured to receive the holding member **112**, the interior member **114**, and part of the thermal block **116**. The heat source **110** is configured to heat the holding and interior members **112**, **114** for heating the filament.

(37) With reference to FIGS. **4**, **5A**, **5B** and **5C**, the holding member **112**, received in the receiving section **122** of the heat source **110**, is configured to hold the interior member **114**, the thermal block **116** and the nozzle **86**. The holding member **112** has an upper connection portion **130** and a lower connection portion **134**, and defines a receiving section **132** therein. In the illustrated embodiment, the upper connection portion **130** is threaded, which enables the holding member **112** to connect with the thermal block **116**. The receiving section **132** is sized to receive the interior member **114**, and the lower connection portion **134** is threaded, which enables the holding member **112** to connect with the nozzle **86**. It is contemplated that in some embodiments, the upper and lower connection portions **130**, **134** could be differently arranged, i.e., not threaded. For example, in some embodiments, the connection between the upper connection **130** and the thermal block **116** could be provided by an interference fit.

(38) The thermal block **116**, disposed at a top of the melt zone component **84**, has a tip portion **140**, an intermediate portion **142** and a connection portion **144**. The thermal block **116** defines a channel **146** that extends along an entire length of the thermal block **116** (i.e., through the tip, intermediate and connection portions **140**, **142**, **144**). The tip portion **140** is partially received in the housing **90** such that filament advanced by the filament advancing members **100** is received into the channel **146**. The intermediate portion **142** is radially larger than the tip portion and sized so as to fully enclose a top of the holding member **112**. This can assist in reducing heat transfer from the melt zone component **84** to the cold end component **82**. The connection portion **144** is complementary to the upper connection portion **130**, such that, in the present embodiment, the connection portion **144**

is threaded. The thermal block **116** can be made of a material having low thermal conductivity and high melting fusion point, such as, for instance, titanium or stainless steel.

(39) The interior member **114** is received in the receiving section **132**, below the thermal block **116**, and will be described in greater detail below.

(40) The nozzle **86** is connected to the holding member **112**, below and downstream of the interior member **114**. It is to be noted that when the heat block **116** and the nozzle **86** are connected to the holding member **112**, the interior member **114** is kept from moving along a length of the holding member **112**.

(41) The nozzle **86** has a connection portion **150** and a tip portion **152**. The connection portion **150** is complementary to the lower connection portion **134**, in the present embodiment, the connection portion **150** being threaded. A receiving channel **154** is defined in the nozzle **86**, and is configured to receive the filament that has been heated by the melt zone component **84**, i.e., the nozzle **86** is fluidly connected to the interior member **114**. Additionally, the tip portion **152** defines an output aperture **156** that is configured to output the heated filament. It is contemplated that the shape of the receiving channel **154** could vary from one embodiment to another. For example, in some instances, the receiving channel **154** could define a sharp conical shape for better guiding the heated filament exiting from the melt interior member **114** to the output aperture **156**. The size and shape of the output aperture **156** could also vary in different embodiments.

(42) Referring to FIGS. 5B, 5C, 6, 7A, 7B and 7C, the interior member **114** will now be described in greater detail. The interior member **114**, which has a generally cylindrical form, has a top end **160** and a bottom end **162**. When the interior member **114** is received in the receiving section **132** of the holding member **112**, the top end **160** is proximate to the thermal block **116**, and the bottom end **162** is proximate to the nozzle **86**.

(43) The interior member **114** defines a passage **170**, three melt channels **172a**, **172b**, **172c**, three upper apertures **174a**, **174b**, **174c**, three intermediate apertures **176a**, **176b**, **176c**, and three lower apertures **178a**, **178b**, **178c**. As will be described below, the number of melt channels and/or the number of apertures could change from one embodiment to another without departing from the scope of the present technology.

(44) Broadly, the passage **170** is fluidly connected to the melt channels **172a**, **172b**, **172c**. More specifically, the passage **170** is fluidly connected to the melt channel **172a** by the upper, intermediate and lower apertures **174a**, **176a**, **178a**; the passage **170** is fluidly connected to the melt channel **172b** by the upper, intermediate and lower apertures **174b**, **176b**, **178b**; and the passage **170** is fluidly connected to the melt channel **172c** by the upper, intermediate and lower apertures **174c**, **176c**, **178c**. One will note that the melt channels **172a**, **172b**, **172c** are also fluidly connected to one another via the apertures. As will be described in greater detail below, the connection configuration between the passage **170** and the melt channels **172a**, **172b**, **172c** could vary from one embodiment to another.

(45) The passage **170** is defined from the top end **160** to the bottom end **162** of the interior member **114**. As will be described below, it is contemplated that in some embodiments, the passage **170** could be defined to extend along only a portion of the length of the interior member **114** (i.e., not from the top end **160** to the bottom end **162**). The channel **170** defines a central channel axis **180**, which, as shown in the accompanying Figures, extends along a length of the passage **170**, and through a radial center thereof. The passage **170** has an entry section **182a**, an upper section **182b**, an intermediate section **182c**, and a bottom section **182d**.

(46) The entry section **182a** extends from the top end **160** to the upper apertures **174a**, **174b**, **174c**, the upper section **182b** extends from the upper apertures **174a**, **174b**, **174c** to the intermediate apertures **176a**, **176b**, **176c**, the intermediate section **182c** extends from the intermediate apertures **176a**, **176b**, **176c** to the lower apertures **178a**, **178b**, **178c**, and the lower section **182d** extends from the lower apertures **178a**, **178b**, **178c** to the bottom end **162**.

(47) The entry section **182a** defines an entry diameter D.sub.E, the upper section **182b** defines an

upper diameter D.sub.U, the intermediate section **182c** defines an intermediate diameter D.sub.I, and the lower section **182d** defines a lower diameter D.sub.L. In the illustrated embodiment, the diameter of the passage **170** decreases from the top end **160** to the bottom end **162**. More specifically, the entry diameter D.sub.E is greater than the upper diameter D.sub.U, the intermediate diameter D.sub.I and the lower diameter D.sub.L. The upper diameter D.sub.U is also greater than the intermediate and lower diameters D.sub.I, D.sub.L, and the intermediate diameter D.sub.I is greater than the lower diameter D.sub.L. It is contemplated that in some embodiments, the diameter of the passage **170** could be uniform along the length of thereof. In other embodiments, the diameter of the passage **170** could gradually change along the length of the interior member **114**, such that the passage **170** could define a frustoconical shape, for example. (48) As best seen in FIGS. **6** and **7B**, the three melt channels **172a**, **172b**, **172c**, are equally angularly spaced from one another. The melt channels **172a**, **172b**, **172c** being angularly spaced from one another provide a symmetrical shape to the interior member **114**, which can assist in uniformizing heat transfer within the interior member **114**.

(49) The melt channel **172a** defines a melt channel axis **190a** extending along a length of the melt channel **172a**, the melt channel **172b** defines a melt channel axis **190b** extending along a length of the melt channel **172b**, and the melt channel **172c** defines a melt channel axis **190c** extending along a length of the melt channel **172c**. The melt channels **172a**, **172b**, **172c** are oriented such that the central melt axes **190a**, **190b**, **190c** are generally parallel to one another, and are generally parallel to the central passage axis **180**. It is contemplated that in some embodiments, the melt channels **172a**, **172b**, **172c** could be oriented such that the central melt axis **190a**, **190b**, **190c** are skewed relative to the central passage axis **180**.

(50) Referring to FIGS. **7A** and **7C**, being that the melt channels **172a**, **172b**, **172c** are similar, only the melt channel **172a** will be described in detail herebelow.

(51) The melt channel **172a** is defined along part of the length of the interior member **114**. While referred to as a “channel” herein, it is noted that the melt channel **172a** is formed by an exterior form of the interior member **114**, but forms a passage through the melt zone component **84** between the interior member **114** and the holding member **112**. In the illustrated embodiment, the melt channel **172a** is defined from the upper aperture **174a** to the bottom end **162**. It is contemplated that the melt channel **172a** could be defined differently. For example, in some instance, the melt channel **172a** could be defined from the upper aperture **174a** to the lower aperture **178a**.

Additionally, the melt channel **172a** has an arcuate cross-sectional form. It is contemplated that the form of the melt channel **172a** could vary from one embodiment to another. For example, in some embodiments, the melt channel **172a** could have a triangular or a rectangular cross-sectional shape. The melt channel **172a** has an upper section **192a**, an intermediate section **192b**, and a lower section **192c**.

(52) The upper section **192a** extends from the upper aperture **174a** to the intermediate aperture **176a**, the intermediate section **192b** extends from the intermediate aperture **176a** to the lower aperture **178a**, and the lower section **192c** extends from the lower aperture **178a** to the bottom end **162** of the interior member **114**.

(53) The melt channel **172a** is configured such that a size thereof increases along the length of the interior member **114**. In the illustrated embodiment, the size of the melt channel **172a** increases from one section to another. More specifically, in the illustrated embodiment, the upper section **192a** has an upper radial depth RD.sub.U measured between an innermost radial point thereof and an outermost radial point thereof. The outermost radial point is coincident with a projection axis **194** (shown in dotted lines in FIG. **7C**). The intermediate section **192b** has an intermediate radial depth RD.sub.I measured between the innermost radial point thereof and the outermost radial point thereof (which is also coincident with the projection axis **194**). The lower section **192c** has a lower radial depth RD.sub.L measured between an innermost radial point thereof and the outermost radial point thereof (which is also coincident with the projection axis **194**). The upper radial depth

RD.sub.U is greater than the intermediate and lower radial depths RD.sub.I, RD.sub.L, and the intermediate radial depth RD.sub.I is greater than the lower radial RD.sub.L. In other words, the depth of the melt channel **172a** varies along the length of the melt channel **172a**.

(54) It is contemplated that in other embodiments, the size of the melt channel **172a** could vary differently. For example, in some embodiments, the size of the upper, intermediate and lower sections **192a**, **192b**, **192c** could vary in an angular fashion, with the upper section **192a** having the smallest angular size, and the lower section **192c** having the largest angular size.

(55) With continued reference to FIGS. **5C**, **6**, **7A** and **7C**, the melt channel **172a** is fluidly connected to the passage **170** by the upper, intermediate and lower apertures **174a**, **176a**, **178a**. It is to be noted that the other upper, intermediate and lower apertures **174b**, **174c**, **176b**, **176c**, **178b**, **178c** are similar to the upper, intermediate and lower apertures **174a**, **176a**, **178a**. It is contemplated that the interior member **114** could define a different number of apertures. For example, referring to FIG. **9**, which depicts an alternative embodiment of the interior member **114**, namely interior member **114'**, the interior member **114'** defines only two apertures **174'**, **176'** per melt channel. In yet other embodiments, the interior member **114** could define a single aperture per melt channel. In some other embodiments, there could be a single aperture connecting two melt channels to the passage **170**.

(56) Returning to **5C**, **6**, **7A** and **7C**, the upper aperture **174a** defines a central aperture axis **196a** that is generally perpendicular to a plane containing the upper aperture **174a**, the intermediate aperture **176a** defines a central aperture axis **196b** that is generally perpendicular to a plane containing the intermediate aperture **176a**, and the lower aperture **178a** defines a central aperture axis **196c** that is generally perpendicular to a plane containing the lower aperture **178a**. In the present embodiment, the respective planes of the upper, intermediate and lower apertures **174a**, **176a**, **178a** are generally parallel to the central passage axis **180**, such that the central aperture axes **196a**, **196b**, **196c** are generally perpendicular to the central passage axis **180**. As will be described in greater detail below, in other embodiments, one or more of the central aperture axes **196a**, **196b**, **196c** could be skewed relative to the central passage axis **180**.

(57) The upper aperture **174a** defines an upper aperture diameter D.sub.UA, the intermediate aperture **176a** defines an intermediate aperture diameter D.sub.IA, and the lower aperture **178a** defines a lower aperture diameter D.sub.LA. The upper aperture diameter D.sub.UA is greater than the intermediate and lower aperture diameters D.sub.IA, D.sub.LA, and the intermediate aperture diameter D.sub.IA is greater than the lower aperture diameter D.sub.LA.

(58) With reference to FIG. **10A**, **10B**, **10C**, another non-limiting embodiment of the interior member **114**, namely interior member **114''**, is illustrated. Features of the interior member **114''** similar to those of the interior member **114** have been labeled with the same reference numerals and will not be described in detail again.

(59) The interior member **114''** defines four melt channels **172a**, **172b**, **172c**, **172d**. The four melt channels **172a**, **172b**, **172c**, **172d** are equally angularly spaced. Additionally, the interior member **114''** defines four upper apertures **174**, four intermediate apertures **176** and four lower apertures **178**.

(60) The passage **170** of the interior member **114''** is defined from the top end **160** of the interior member **114** to the lower apertures **178** and does not extend to a bottom end thereof. Thus, the passage **170** and the melt channels **172a**, **172b**, **172c**, **172d** merge upstream from the bottom end **162**. In the present embodiment, the central aperture axes of the apertures **174**, **176**, **178** (only two apertures **178** being shown in FIG. **10C**) are skewed relative to the central passage axis **180**, as is illustrated by axes **196**.

(61) Referring back to FIGS. **3**, **4**, **5A** to **5C**, **6** and **7A** to **7C**, the operation of the extruder head assembly **56** will now be described.

(62) To begin, the heater **120** of the heat source **110** is turned on. The heat source **110** heats the holding member **112**, the interior member **114**, and the nozzle **86** until a predetermined temperature

is reached. In some instances, the predetermined temperature is greater than the melting temperature of the filament. In other instances, the predetermined temperature could be below the melting temperature of the filament. As the effective melting temperature of different filaments could vary depending on base material, additives, etc., the pre-determined temperature may not in some cases be specifically tuned to the melting temperature of the filament used. It is to be noted that the thermal block **116**, while also being heated by the heat source **110**, assists in reducing heat transfer from the heat source **110**, the holding member **112** and the interior member **114** to the cold end component **82** (in part due to its low thermal conductivity).

(63) During operation, filament is provided to the extruder head assembly **56**. Filament that is received into the cold end component **82** through the feeding tip **96** and is engaged between the advancing members **100a**, **100b**. Rotation of the advancing members **100a**, **100b** causes the filament to move along the housing **90** and eventually be received into the melt zone component **84**. It is to be noted that during operation, the advancing members **100a**, **100b** are continuously rotated, and as such, filament is continuously fed into the melt zone component **84**, barring any slippage or blocked of the filament.

(64) The filament enters the melt zone component **84** by being received into the channel **146** of the thermal block **116** and entering into the entry section **182a** of the passage **170**.

(65) Once the filament enters the passage **170**, there is heat transfer from the interior member **114** to the filament and the filament begins to melt. However, it is to be noted that due to the heating configuration, the filament does not heat up and/or melt equally: a radially outer surface of the filament is at a higher temperature than a core of the filament. It is contemplated that in some embodiments, the filament could begin to melt in the thermal block **116**.

(66) Since the advancing members **100a**, **100b** are continuously rotating, the filament is continuously advancing. While advancing along the entry section **182a**, part of the filament continues to melt. Eventually, the upper apertures **174a**, **174b**, **174c** are reached. At this stage, some of the filament has turned into melted filament material. The melted filament material flows into the melt channels **172a**, **172b**, **172c** through the upper apertures **174a**, **174b**, **174c**. It is to be noted that in some embodiments where the upper apertures **174a**, **174b**, **174c** are skewed, the melted filament material may be guided into the melt channels **172a**, **172b**, **172c** more easily.

(67) The unmelted filament flows past the upper apertures **174a**, **174b**, **174c** and enters the upper section **182b**. Since some of the filament has melted, the diameter of the unmelted filament entering the upper section **182b** is less than the diameter of the filament entering the entry section **182a**. As such, the upper diameter $D_{sub.U}$ being smaller than the entry diameter $D_{sub.E}$ can assist in minimizing clearance between the unmelted filament and the interior member **114**, which can enhance heat transfer therebetween. Additionally, in some cases, the upper diameter $D_{sub.U}$ has been reduced such that only the unmelted filament material can be received therein, thereby encouraging the melted filament material to flow into the melt channels **172a**, **172b**, **172c**, **172d**.

(68) While advancing along the upper section **182b**, the filament continues to melt. When arriving at the intermediate apertures **176a**, **176b**, **176c**, more of the filament has melted. The melted filament material flows into the melt channels **172a**, **172b**, **172c** through the intermediate apertures **176a**, **176b**, **176c**.

(69) As described above, the unmelted filament flows past the intermediate apertures **176a**, **176b**, **176c**, and enters the intermediate section **182c**. Since some of the filament has further melted, the diameter of the unmelted filament entering the intermediate section **182c** is less than the diameter of the filament entering the upper section **182b**. Thus, as described above, reduction of the intermediate diameter $D_{sub.I}$, can assist in heat transfer.

(70) Similar to the above, the filament continues to flow in the passage **170** until the bottom end **162** is reached. At the bottom end **162**, a core of the filament is meant to have reached the desired temperature and melted as well.

(71) As the filament is flowing in the passage **170** toward the lower end **162**, the melted filament

material is also flowing in the melt channels **172a**, **172b**, **172c** toward the lower end **162**. It is to be noted that the melted filament material can, in some instances, engage an inner surface of the holding member **112** while moving toward the lower end **162**. Additionally, a temperature and/or the melted state of the melted filament material should generally be maintained, as the melted filament material is also subject to heat transfer from the holding and interior members **112**, **114**. (72) Finally, the output from the passage **170** and the melt channels **172a**, **172b**, **172c** merge into the nozzle **86**, before being extruded through the output aperture **156**. Upon combination in the nozzle **86**, all extruded filament material is meant to have melted and be at a desired temperature. In at least some embodiments, recombination of the passage **170** and the melt channels **172a**, **172b**, **172c** could take place in a lower portion of the interior member **114**, prior to arrival at the nozzle **86**.

(73) Modifications and improvements to the above-described embodiments of the present invention may become apparent to those skilled in the art. The foregoing description is intended to be exemplary rather than limiting. The scope of the present invention is therefore intended to be limited solely by the appended claims.

Claims

1. An extruder head assembly for an additive manufacturing machine, the extruder head assembly comprising: a melt zone component operatively connected to a cold end component for receiving and heating a filament advanced by the cold end component; and a nozzle operatively connected to the melt zone component, the nozzle being configured to receive the filament having been heated from the melt zone component, the melt zone component comprising: a heat source, and an interior member in thermal communication with the heat source, the interior member defining therein: a passage configured to heat unmelted material of the filament, at least one melt channel, and at least one aperture fluidly connecting the passage to the at least one melt channel, the at least one aperture being configured to direct melted filament material toward the at least one melt channel, the passage and the at least one melt channel fluidly connecting downstream from the at least one aperture, the nozzle being configured to receive heated filament material from the passage and the at least one melt channel.
2. The extruder head assembly of claim 1, wherein: the at least one aperture defines a central aperture axis that is generally perpendicular to a plane containing the at least one aperture, the passage defines a central passage axis extending along a length of the passage, and the central aperture axis is one of: generally perpendicular to the central passage axis; and generally skewed relative to the central passage axis.
3. The extruder head assembly of claim 2, wherein: the at least one melt channel defines a central melt axis extending along the length of the at least one melt channel; and the central melt axis is parallel to the central passage axis.
4. The extruder head assembly of claim 1, wherein: the passage defines a central passage axis extending along a length of the passage; the at least one melt channel defines a melt channel axis extending along a length of the at least one melt channel; and the melt channel axis is parallel to the central passage axis.
5. The extruder head assembly of claim 1, wherein the passage has: a first passage diameter upstream from the at least one aperture; a second passage diameter downstream from the at least one aperture; and the first passage diameter is greater than the second passage diameter.
6. The extruder head assembly of claim 1, wherein the at least one aperture includes a first aperture and a second aperture, the second aperture being spaced from the first aperture along a length of the interior member.
7. The extruder head assembly of claim 6, wherein the first aperture has a first aperture diameter, the second aperture has a second aperture diameter, and the first aperture diameter is greater than

the second aperture diameter.

8. The extruder head assembly of claim 6, wherein the passage has: a first passage diameter upstream from the first aperture; a second passage diameter downstream from the first aperture and upstream from the second aperture; and a third passage diameter downstream from the second aperture, and the first passage diameter is greater than the second and third passage diameters, and the second passage diameter is greater than the third passage diameter.

9. The extruder head assembly of claim 1, wherein the at least one melt channel includes a first melt channel and a second melt channel.

10. The extruder head assembly of claim 9, wherein the at least one aperture fluidly connects the passage to the first and second melt channels.

11. The extruder head assembly of claim 9, wherein the at least one aperture includes a first aperture fluidly connecting the passage to the first melt channel, and a second aperture fluidly connecting the passage to the second melt channel.

12. The extruder head assembly of claim 1, wherein: the at least one melt channel includes a first melt channel, a second melt channel and a third melt channel; and the at least one aperture includes: a first aperture fluidly connecting the passage to the first melt channel, a second aperture fluidly connecting the passage to the second melt channel, and a third aperture fluidly connecting the passage to the third melt channel.

13. The extruder head assembly of claim 12, further including a fourth aperture fluidly connecting the passage to the first melt channel, the fourth aperture being spaced from the first aperture along a length of the interior member.

14. The extruder head assembly of claim 13, wherein the first aperture has a first aperture diameter, the fourth aperture has a second aperture diameter, and the first aperture diameter is greater than the second aperture diameter.

15. The extruder head assembly of claim 13, wherein the passage has: a first passage diameter upstream from the first aperture; a second passage diameter downstream from the first aperture and upstream from the fourth aperture; and a third passage diameter downstream from the fourth aperture, and the first passage diameter is greater than the second and third passage diameters, and the second passage diameter is greater than the third passage diameter.

16. The extruder head assembly of claim 1, wherein the melt zone component further includes a holding member selectively receiving the interior member therein, one end of the holding member being selectively connected to the cold end component, and another end of the holding member being selectively connected to the nozzle.

17. The extruder head assembly of claim 1, further comprising the cold end component for receiving filament, the cold end component including filament advancing members.
